

CBSE Class 12 Maths — Half-Yearly Predicted Paper (Set 7 of 10)

Based on the latest CBSE 2026-27 competency-based blueprint and highly-repeated past year questions (PYQs), here is Set 7 of the Half-Yearly Predicted Paper. This set completely excludes Linear Programming and Probability, focusing rigorously on the ~80% half-yearly syllabus. Every question here is brand-new compared to the earlier sets.

SECTION A: Objective Type Questions (1 Mark Each x 20 = 20 Marks)

- Q1.** Let the function $f: \mathbb{R} - \{\frac{2}{3}\} \rightarrow \mathbb{R} - \{\frac{2}{3}\}$ be defined by $f(x) = \frac{4x+3}{6x-4}$. Find the composition $f(f(x))$. * **Chapter:** Relations and Functions * **Confidence: High**
- Q2.** Find the principal value of $\sin^{-1}(\sin \frac{3\pi}{5})$. * **Chapter:** Inverse Trigonometric Functions * **Confidence: Very High**
- Q3.** If $A = \begin{pmatrix} 0 & a \\ -3 & 0 \end{pmatrix}$ is a skew-symmetric matrix, find the exact value of a . * **Chapter:** Matrices * **Confidence: Very High**
- Q4.** If A is a square matrix of order 3×3 such that its determinant $|A| = 5$, find the value of $|2A|$. * **Chapter:** Determinants * **Confidence: High**
- Q5.** Find the derivative of $y = \log_e(\sin x)$ with respect to x . * **Chapter:** Continuity and Differentiability * **Confidence: Very High**
- Q6.** The side of a square is increasing at the constant rate of 0.2 cm/s . Find the rate of increase of its perimeter. * **Chapter:** Application of Derivatives * **Confidence: Medium**
- Q7.** Evaluate the indefinite integral: $\int \frac{dx}{x^2+16}$. * **Chapter:** Integrals * **Confidence: High**
- Q8.** Evaluate the definite integral: $\int_0^{\pi/2} \cos x \, dx$. * **Chapter:** Integrals * **Confidence: Very High**
- Q9.** State the degree of the differential equation $x\left(\frac{dy}{dx}\right)^3 + y\frac{d^2y}{dx^2} = 0$. * **Chapter:** Differential Equations * **Confidence: Very High**
- Q10.** Write the integrating factor (I.F.) for the linear differential equation: $\frac{dy}{dx} + y \cot x = \cos x$. * **Chapter:** Differential Equations * **Confidence: High**
- Q11.** Find the scalar dot product of the vectors $\vec{a} = 2\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} + 4\hat{j} - \hat{k}$. * **Chapter:** Vector Algebra * **Confidence: Very High**
- Q12.** Find a unit vector in the direction of the vector $\vec{a} = \hat{i} + \hat{j} + \sqrt{2}\hat{k}$. * **Chapter:** Vector Algebra * **Confidence: Medium**
- Q13.** Calculate the distance of the point $P(3, 4, 5)$ from the origin. * **Chapter:** Three-Dimensional Geometry * **Confidence: High**
- Q14.** What are the direction cosines of a line which is equally inclined to the negative directions of the coordinate axes? * **Chapter:** Three-Dimensional Geometry * **Confidence: High**
- Q15.** Let R be a relation defined on the set of natural numbers $\mathbb{N}$ by $x + 2y = 8$. Find the domain of the relation R . * **Chapter:** Relations and Functions * **Confidence: Medium**
- Q16.** Construct a 2×2 matrix $A = [a_{ij}]$ whose elements are given by $a_{ij} = \frac{(i+2j)^2}{2}$. Find the specific value of the element a_{21} . * **Chapter:** Matrices * **Confidence: High**
- Q17.** Find the interval in which the function $f(x) = x^2 - 4x + 6$ is strictly decreasing. * **Chapter:** Application of Derivatives * **Confidence: Very High**
- Q18.** Evaluate the indefinite integral: $\int e^x \left(\frac{1}{x} - \frac{1}{x^2}\right) dx$. * **Chapter:** Integrals * **Confidence: Very High**
- Q19. (Assertion-Reason): Assertion (A):** The line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ is strictly perpendicular to the line $\frac{x}{2} = \frac{y}{-4} = \frac{z}{2}$. **Reason (R):** Two lines with direction ratios a_1, b_1, c_1 and a_2, b_2, c_2 are perpendicular if and only if $a_1a_2 + b_1b_2 + c_1c_2 = 0$. Choose the correct option. * **Chapter:** Three-Dimensional Geometry * **Confidence: High**
- Q20. (Assertion-Reason): Assertion (A):** The greatest integer function $f(x) = [x]$ is discontinuous at all integral points. **Reason (R):** At integral points, the left-hand limit is not equal to the right-hand limit for the greatest integer function. Choose the correct option. * **Chapter:** Continuity and Differentiability * **Confidence: Very High**

SECTION B: Very Short Answer Type Questions (2 Marks Each x 5 = 10 Marks)

- Q21.** Prove the algebraic identity: $3 \sin^{-1} x = \sin^{-1}(3x - 4x^3)$, for $x \in [-1/2, 1/2]$. * **Chapter:** Inverse Trigonometric Functions * **Confidence: Very High**
- Q22.** Using the determinant method, find the area of the triangle whose vertices are $(1, 0)$, $(6, 0)$ and $(4, 3)$. * **Chapter:** Determinants * **Confidence: High**
- Q23.** Find $\frac{dy}{dx}$ implicitly if $x^2 + xy + y^2 = 100$. * **Chapter:** Continuity and Differentiability * **Confidence: Very High**
- Q24.** Find a vector in the direction of vector $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ which has a magnitude of exactly 6 units. * **Chapter:** Vector Algebra * **Confidence: High**

Q25. Solve the differential equation by the method of variable separation: $\frac{dy}{dx} = \sin^2 x$. * **Chapter:** Differential Equations * **Confidence: Medium**

SECTION C: Short Answer Type Questions (3 Marks Each x 6 = 18 Marks)

Q26. Show that the relation R in the set $A = \{1, 2, 3, 4, 5\}$ given by $R = \{(a, b) : |a - b| \text{ is a multiple of } 2\}$ is an equivalence relation. * **Chapter:** Relations and Functions * **Confidence: Very High**

Q27. If $A = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$, prove using the principle of mathematical induction that $A^n = \begin{pmatrix} \cos n\theta & \sin n\theta \\ -\sin n\theta & \cos n\theta \end{pmatrix}$ for all $n \in \mathbb{N}$. * **Chapter:** Matrices * **Confidence: High**

Q28. Differentiate the function $y = (\tan x)^{\cot x} + (\cot x)^{\tan x}$ with respect to x . * **Chapter:** Continuity and Differentiability * **Confidence: Very High**

Q29. Evaluate the indefinite integral using the method of partial fractions: $\int \frac{x}{(x-1)(x-2)(x-3)} dx$. * **Chapter:** Integrals * **Confidence: Very High**

Q30. Find the particular solution of the linear differential equation $\frac{dy}{dx} + y \sec x = \tan x$, given that $y = 1$ when $x = 0$. * **Chapter:** Differential Equations * **Confidence: High**

Q31. Find the coordinates of the exact point where the straight line passing through the points $(5, 1, 6)$ and $(3, 4, 1)$ crosses the yz -plane. * **Chapter:** Three-Dimensional Geometry * **Confidence: Very High**

SECTION D: Long Answer Type Questions (5 Marks Each x 4 = 20 Marks)

Q32. Use the matrix inverse method to solve the following system of linear equations: $3x - 2y + 3z = 8$, $2x + y - z = 1$, $4x - 3y + 2z = 4$. * **Chapter:** Matrices and Determinants * **Confidence: Very High**

Q33. A wire of length **36 m** is to be cut into two pieces. One of the pieces is to be bent into the shape of a square and the other into the shape of an equilateral triangle. Using derivatives, find the exact length of each piece so that the combined area of the square and the triangle is mathematically minimum. * **Chapter:** Application of Derivatives * **Confidence: Very High**

Q34. Evaluate the following indefinite integral: $\int (\sqrt{\tan x} + \sqrt{\cot x}) dx$. * **Chapter:** Integrals * **Confidence: High**

Q35. Using the method of definite integration, sketch the region and find the exact area bounded by the curve (parabola) $y = x^2$ and the horizontal line $y = 4$. * **Chapter:** Application of Integrals * **Confidence: Very High**

SECTION E: Case-Study Based Questions (4 Marks Each x 3 = 12 Marks)

Q36. Case Study 1: Matrices (School Merit Awards) Two schools, A and B, want to award their selected students for the core values of sincerity, truthfulness, and helpfulness. School A wants to award ₹ x , ₹ y , and ₹ z for the three values respectively to 3, 2, and 1 students, distributing a total award money of ₹ 1600. School B wants to spend ₹ 2300 to award its 4, 1, and 3 students on the respective values (using the same award amounts). The total amount of the award for one prize on each value is precisely ₹ 900. (i) Formulate this scenario mathematically into a matrix equation $AX = B$. (1 Mark) (ii) Find the determinant $|A|$ of the coefficient matrix. (1 Mark) (iii) Using the matrix inverse method, find the exact award money for sincerity, truthfulness, and helpfulness. (2 Marks) * **Chapter:** Matrices and Determinants * **Confidence: High**

Q37. Case Study 2: Application of Derivatives (Stadium Design) An architect is designing an athletic stadium with a rectangular playing field. To maximize the usable area for sports events, the field must be enclosed by a running track with a fixed inner perimeter of exactly **400 meters**. The track configuration consists of a central rectangle with two perfect semicircular ends. Let the length and width of the rectangular part be x and y respectively. (i) Express the width y purely in terms of the length x and the perimeter. (1 Mark) (ii) Find the area function A of the rectangular field strictly in terms of x . (1 Mark) (iii) Using calculus derivative tests, find the dimensions x and y that will maximize the area of the rectangular playing field. (2 Marks) * **Chapter:** Application of Derivatives * **Confidence: Very High**

Q38. Case Study 3: 3D Geometry (Helicopter & Drone Rescue) During a search and rescue operation in a mountainous region, a rescue helicopter is flying along a linear flight path represented by the vector line $L_1 : \vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$. Simultaneously, an automated supply drone is deployed along another linear flight path $L_2 : \vec{r} = (2\hat{i} - \hat{j} - \hat{k}) + \mu(2\hat{i} + \hat{j} + 2\hat{k})$. To drop supplies safely between the two crafts, the control room needs to find the exact shortest distance between their paths. (i) Identify the direction vectors $\vec{b}_1$ and $\vec{b}_2$ of the two flight paths and calculate their cross product $(\vec{b}_1 \times \vec{b}_2)$. (2 Marks) (ii) Calculate the exact shortest distance between the helicopter and the drone's flight paths. (2 Marks) * **Chapter:** Three-Dimensional Geometry * **Confidence: High**

Answer Key

$$\text{Q1. } f(f(x)) = \frac{4(\frac{4x+3}{6x-4})+3}{6(\frac{4x+3}{6x-4})-4} = \frac{16x+12+18x-12}{24x+18-24x+16} = \frac{34x}{34} = x. \text{ Q2.}$$

$$\sin^{-1}(\sin \frac{3\pi}{5}) = \sin^{-1}(\sin(\pi - \frac{2\pi}{5})) = \sin^{-1}(\sin \frac{2\pi}{5}) = \frac{2\pi}{5}. \text{ Q3. For a skew-symmetric matrix, } a_{ij} = -a_{ji}. \text{ Thus,}$$

$$a_{12} = -a_{21} \Rightarrow a = -(-3) = 3. \text{ Q4. } |2A| = 2^3|A| = 8 \times 5 = 40. \text{ Q5. } \frac{dy}{dx} = \frac{1}{\sin x} \cdot \cos x = \cot x. \text{ Q6. Perimeter}$$

$$P = 4s \Rightarrow \frac{dP}{dt} = 4 \frac{ds}{dt} = 4(0.2) = 0.8 \text{ cm/s. Q7. } \int \frac{dx}{x^2+4} = \frac{1}{4} \tan^{-1}(\frac{x}{2}) + C. \text{ Q8.}$$

$$[\sin x]_0^{\pi/2} = \sin(\pi/2) - \sin(0) = 1 - 0 = 1. \text{ Q9. The highest order derivative is } \frac{d^2y}{dx^2}. \text{ Its power is 1, so the degree is 1. Q10.}$$

$$\text{I.F.} = e^{\int \cot x dx} = e^{\log |\sin x|} = \sin x. \text{ Q11. } \vec{a} \cdot \vec{b} = (2)(1) + (-1)(4) + (3)(-1) = 2 - 4 - 3 = -5. \text{ Q12. Magnitude}$$

$$= \sqrt{1^2 + 1^2 + (\sqrt{2})^2} = \sqrt{1+1+2} = \sqrt{4} = 2. \text{ Unit vector} = \frac{1}{2}\hat{i} + \frac{1}{2}\hat{j} + \frac{\sqrt{2}}{2}\hat{k}. \text{ Q13. Distance}$$

$$= \sqrt{3^2 + 4^2 + 5^2} = \sqrt{9+16+25} = \sqrt{50} = 5\sqrt{2} \text{ units. Q14. Let DCs be } -l, -l, -l \text{ (since they are equal and negative).}$$

$$(-l)^2 + (-l)^2 + (-l)^2 = 1 \Rightarrow 3l^2 = 1 \Rightarrow l = \frac{1}{\sqrt{3}}. \text{ DCs are } -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}. \text{ Q15. } x + 2y = 8 \Rightarrow x = 8 - 2y$$

$$\text{. For } x, y \in \mathbb{N}, \text{ possible values for } y \text{ are } 1, 2, 3 \text{ (giving } x = 6, 4, 2). \text{ Domain (values of } x) \text{ is } \{2, 4, 6\}. \text{ Q16.}$$

$$a_{21} = \frac{(2+2(1))^2}{2} = \frac{4^2}{2} = \frac{16}{2} = 8. \text{ Q17. } f'(x) = 2x - 4. \text{ For strictly decreasing, } f'(x) < 0 \Rightarrow 2x - 4 < 0 \Rightarrow x < 2.$$

$$\text{Interval is } (-\infty, 2). \text{ Q18. Form is } \int e^x(f(x) + f'(x))dx \text{ where } f(x) = \frac{1}{x} \text{ and } f'(x) = -\frac{1}{x^2}. \text{ Answer is } e^x \cdot \frac{1}{x} + C. \text{ Q19.}$$

$$\text{Option A. (Both A and R are true, and R is the correct explanation. } 2(2) + 3(-4) + 4(2) = 4 - 12 + 8 = 0). \text{ Q20. Option A.}$$

$$\text{(Both A and R are true, and R is the correct explanation of A).}$$

$$\text{Q21. Let } x = \sin \theta \Rightarrow \theta = \sin^{-1} x. \text{ RHS} = \sin^{-1}(3 \sin \theta - 4 \sin^3 \theta) = \sin^{-1}(\sin 3\theta) = 3\theta = 3 \sin^{-1} x = \text{LHS. Q22.}$$

$$\text{Area} = \frac{1}{2} \begin{vmatrix} 1 & 0 & 1 \\ 6 & 0 & 1 \\ 4 & 3 & 1 \end{vmatrix} = \frac{1}{2} |1(0-3) - 0 + 1(18-0)| = \frac{1}{2} |-3 + 18| = \frac{15}{2} = 7.5 \text{ sq units. Q23. Differentiating both}$$

$$\text{sides: } 2x + (x \frac{dy}{dx} + y) + 2y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx}(x + 2y) = -(2x + y) \Rightarrow \frac{dy}{dx} = -\frac{2x+y}{x+2y}. \text{ Q24.}$$

$$|\vec{a}| = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{9} = 3. \text{ Unit vector } \hat{a} = \frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} + \frac{2}{3}\hat{k}. \text{ Required vector} = 6\hat{a} = 4\hat{i} - 2\hat{j} + 4\hat{k}. \text{ Q25.}$$

$$dy = \sin^2 x dx \Rightarrow \int dy = \int \frac{1-\cos 2x}{2} dx \Rightarrow y = \frac{1}{2}(x - \frac{\sin 2x}{2}) + C.$$

$$\text{Q26. Reflexive: } |a - a| = 0, \text{ which is a multiple of 2. Symmetric: If } |a - b| \text{ is a multiple of 2, then } |-(b - a)| = |b - a| \text{ is also a multiple of 2. Transitive: If } |a - b| \text{ and } |b - c| \text{ are multiples of 2, then } (a - b) \text{ and } (b - c) \text{ are even integers. Their sum } (a - c) = (a - b) + (b - c) \text{ is even, so } |a - c| \text{ is a multiple of 2. Hence, equivalence relation. Q27. Base case } n = 1:$$

$$A^1 = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}, \text{ which is true. Assume true for } n = k: A^k = \begin{pmatrix} \cos k\theta & \sin k\theta \\ -\sin k\theta & \cos k\theta \end{pmatrix}. \text{ Prove for } n = k + 1:$$

$$A^{k+1} = A^k A = \begin{pmatrix} \cos k\theta & \sin k\theta \\ -\sin k\theta & \cos k\theta \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} = \begin{pmatrix} \cos k\theta \cos \theta - \sin k\theta \sin \theta & \cos k\theta \sin \theta + \sin k\theta \cos \theta \\ -\sin k\theta \cos \theta - \cos k\theta \sin \theta & -\sin k\theta \sin \theta + \cos k\theta \cos \theta \end{pmatrix} = \begin{pmatrix} \cos (k+1)\theta & \sin (k+1)\theta \\ -\sin (k+1)\theta & \cos (k+1)\theta \end{pmatrix}$$

$$\text{. Hence proved. Q28. Let}$$

$$u = (\tan x)^{\cot x} \Rightarrow \ln u = \cot x \ln(\tan x) \Rightarrow \frac{1}{u} \frac{du}{dx} = -\operatorname{cosec}^2 x \ln(\tan x) + \cot x \frac{\sec^2 x}{\tan x} = \operatorname{cosec}^2 x(1 - \ln(\tan x))$$

$$\text{. Thus } u' = (\tan x)^{\cot x} \operatorname{cosec}^2 x(1 - \ln(\tan x)). \text{ Let}$$

$$v = (\cot x)^{\tan x} \Rightarrow \ln v = \tan x \ln(\cot x) \Rightarrow v' = (\cot x)^{\tan x} \sec^2 x(1 - \ln(\cot x)). \text{ Add } u' \text{ and } v' \text{ for } \frac{dy}{dx}. \text{ Q29.}$$

$$\frac{x}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}. \text{ Using limits or equating coefficients: } A = \frac{1}{(-1)(-2)} = \frac{1}{2}, B = \frac{2}{(1)(-1)} = -2,$$

$$C = \frac{3}{(2)(1)} = \frac{3}{2}. \text{ Integral becomes } \frac{1}{2} \log |x-1| - 2 \log |x-2| + \frac{3}{2} \log |x-3| + C. \text{ Q30. I.F.}$$

$$= e^{\int \sec x dx} = e^{\log |\sec x + \tan x|} = \sec x + \tan x. \text{ Solution is}$$

$$y(\sec x + \tan x) = \int \tan x(\sec x + \tan x) dx = \int (\sec x \tan x + \sec^2 x - 1) dx = \sec x + \tan x - x + C. \text{ At}$$

$$x = 0, y = 1 \Rightarrow 1(1+0) = 1+0-0+C \Rightarrow C = 0. \text{ Particular solution: } y(\sec x + \tan x) = \sec x + \tan x - x.$$

$$\text{Q31. Line equation passing through } A(5, 1, 6) \text{ and } B(3, 4, 1): \frac{x-5}{-2} = \frac{y-1}{3} = \frac{z-6}{-5} = k. \text{ Any point is}$$

$$(-2k+5, 3k+1, -5k+6). \text{ At the } yz\text{-plane, } x = 0 \Rightarrow -2k+5 = 0 \Rightarrow k = \frac{5}{2}. \text{ The } y\text{-coordinate is}$$

$$3(5/2)+1 = \frac{17}{2}. \text{ The } z\text{-coordinate is } -5(5/2)+6 = -\frac{13}{2}. \text{ Point is } (0, \frac{17}{2}, -\frac{13}{2}).$$

$$\text{Q32. } |A| = 3(2-3) - (-2)(-4+4) + 3(-6-4) = -3-0-30 = -33 \neq 0. \text{ Find cofactor matrix and take its}$$

$$\text{transpose to get } \operatorname{adj}(A) = \begin{pmatrix} -1 & 5 & -1 \\ -8 & -6 & 9 \\ -10 & -1 & 7 \end{pmatrix}. A^{-1} = -\frac{1}{33} \operatorname{adj}(A).$$

$$X = A^{-1}B = -\frac{1}{33} \begin{pmatrix} -1 & 5 & -1 \\ -8 & -6 & 9 \\ -10 & -1 & 7 \end{pmatrix} \begin{pmatrix} 8 \\ 1 \\ 4 \end{pmatrix} = -\frac{1}{33} \begin{pmatrix} -8+5-4 \\ -64-6+36 \\ -80-1+28 \end{pmatrix} = -\frac{1}{33} \begin{pmatrix} -7 \\ -34 \\ -53 \end{pmatrix}. \text{ Thus } x = \frac{7}{33}, y = \frac{34}{33}, z = \frac{53}{33}$$

$$\text{. Q33. Let the square piece length be } x. \text{ Triangle piece length is } 36 - x. \text{ Side of square is } x/4. \text{ Side of triangle is } (36-x)/3.$$

$$\text{Total Area } A = (\frac{x}{4})^2 + \frac{\sqrt{3}}{4}(\frac{36-x}{3})^2 = \frac{x^2}{16} + \frac{\sqrt{3}(36-x)^2}{36}. \text{ Differentiating: } A'(x) = \frac{2x}{16} - \frac{2\sqrt{3}(36-x)}{36} = \frac{x}{8} - \frac{\sqrt{3}(36-x)}{18}. \text{ Set to}$$

$$0: \frac{x}{8} = \frac{\sqrt{3}(36-x)}{18} \Rightarrow 18x = 8\sqrt{3}(36-x) \Rightarrow x = \frac{288\sqrt{3}}{18+8\sqrt{3}} = \frac{144\sqrt{3}}{9+4\sqrt{3}} \text{ m. Square piece is } \frac{144\sqrt{3}}{9+4\sqrt{3}} \text{ m, triangle piece is}$$

$$36 - x \text{ m. } A''(x) > 0 \text{ verifies minimum. Q34.}$$

$$\int \frac{\sqrt{\sin x}}{\sqrt{\cos x}} + \frac{\sqrt{\cos x}}{\sqrt{\sin x}} dx = \int \frac{\sin x + \cos x}{\sqrt{\sin x \cos x}} dx = \sqrt{2} \int \frac{\sin x + \cos x}{\sqrt{2 \sin x \cos x}} dx = \sqrt{2} \int \frac{\sin x + \cos x}{\sqrt{1-(\sin x - \cos x)^2}} dx.$$

$$\text{Let } \sin x - \cos x = t \Rightarrow (\cos x + \sin x) dx = dt. \text{ Integral is}$$

$$\sqrt{2} \int \frac{dt}{\sqrt{1-t^2}} = \sqrt{2} \sin^{-1} t + C = \sqrt{2} \sin^{-1}(\sin x - \cos x) + C. \text{ Q35. The curve } y = x^2 \text{ is an upward opening parabola.}$$

$$\text{Line } y = 4 \text{ is a horizontal line. Intersection points are } x^2 = 4 \Rightarrow x = \pm 2. \text{ Due to symmetry, Area}$$

$$= 2 \int_0^2 (\text{Upper} - \text{Lower}) dx = 2 \int_0^2 (4 - x^2) dx = 2[4x - \frac{x^3}{3}]_0^2 = 2(8 - \frac{8}{3}) = 2(\frac{16}{3}) = \frac{32}{3} \text{ sq units.}$$

Q36. (i) $\begin{pmatrix} 3 & 2 & 1 \\ 4 & 1 & 3 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1600 \\ 2300 \\ 900 \end{pmatrix}$. (ii) $|A| = 3(1-3) - 2(4-3) + 1(4-1) = -6 - 2 + 3 = -5$. (iii) Using inverse $A^{-1} = \frac{1}{-5} \begin{pmatrix} -2 & -1 & 5 \\ -1 & 2 & -5 \\ 3 & -1 & -5 \end{pmatrix}$. $X = A^{-1}B = \frac{1}{-5} \begin{pmatrix} -3200 - 2300 + 4500 \\ -1600 + 4600 - 4500 \\ 4800 - 2300 - 4500 \end{pmatrix} = \frac{1}{-5} \begin{pmatrix} -1000 \\ -1500 \\ -2000 \end{pmatrix} = \begin{pmatrix} 200 \\ 300 \\ 400 \end{pmatrix}$. ₹200 for Sincerity, ₹300 for Truthfulness, ₹400 for Helpfulness. **Q37.** (i) Track perimeter $P = 2x + \pi y = 400 \Rightarrow y = \frac{400-2x}{\pi}$. (ii) Area of rectangular field $A(x) = xy = x \left(\frac{400-2x}{\pi} \right) = \frac{400x-2x^2}{\pi}$. (iii) $A'(x) = \frac{400-4x}{\pi}$. Set $A'(x) = 0 \Rightarrow x = 100$ m. Using (i), $y = \frac{400-200}{\pi} = \frac{200}{\pi}$ m. $A''(x) = -\frac{4}{\pi} < 0$, ensuring maximum area. **Q38.** (i) $\vec{b}_1 = \hat{i} - \hat{j} + \hat{k}$. $\vec{b}_2 = 2\hat{i} + \hat{j} + 2\hat{k}$. Cross product $\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 2 & 1 & 2 \end{vmatrix} = \hat{i}(-2-1) - \hat{j}(2-2) + \hat{k}(1+2) = -3\hat{i} + 0\hat{j} + 3\hat{k}$. (ii) $\vec{a}_2 - \vec{a}_1 = (2-1)\hat{i} + (-1-2)\hat{j} + (-1-3)\hat{k} = \hat{i} - 3\hat{j} - 4\hat{k}$. Shortest Distance $SD = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|} = \frac{|(1)(-3) + (-3)(0) + (-4)(3)|}{\sqrt{9+0+9}} = \frac{|-3-12|}{\sqrt{18}} = \frac{15}{3\sqrt{2}} = \frac{5}{\sqrt{2}} = \frac{5\sqrt{2}}{2}$ units.

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